Computer Networks

4th Semester, Academic Year 2026

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Deliverables

I. ipconfig

```
TANMAYI_PES1UG24CS493>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8382:c808:3b7b:80ff%40
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : PESITWC.EDU
   Link-local IPv6 Address . . . . . : fe80::37c7:8442:7d13:2d6d%6
   IPv4 Address. . . . . . . . . . . : 10.20.201.131
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 10.20.200.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

II. ping

```
TANMAYI_PES1UG24CS493>ping google.com

Pinging google.com [142.250.195.46] with 32 bytes of data:
Reply from 142.250.195.46: bytes=32 time=42ms TTL=113
Reply from 142.250.195.46: bytes=32 time=277ms TTL=113
Reply from 142.250.195.46: bytes=32 time=27ms TTL=113
Reply from 142.250.195.46: bytes=32 time=34ms TTL=113

Ping statistics for 142.250.195.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 27ms, Maximum = 277ms, Average = 95ms
```

III. netstat

```
TANMAYI_PES1UG24CS493>netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49670          0.0.0.0:0              LISTENING
  TCP    10.20.201.131:139      0.0.0.0:0              LISTENING
  TCP    10.20.201.131:49183    172.217.70.188:443     ESTABLISHED
  TCP    10.20.201.131:49186    52.104.12.55:443       TIME_WAIT
  TCP    10.20.201.131:49187    40.104.66.210:443      TIME_WAIT
  TCP    10.20.201.131:49417    4.213.25.242:443       ESTABLISHED
  TCP    10.20.201.131:49671    142.250.71.99:443      ESTABLISHED
  TCP    10.20.201.131:49672    142.250.207.238:443    ESTABLISHED
  TCP    10.20.201.131:49844    142.250.195.46:443     ESTABLISHED
  TCP    10.20.201.131:51435    52.123.171.38:443      ESTABLISHED
  TCP    10.20.201.131:52137    142.251.220.4:443      ESTABLISHED
  TCP    10.20.201.131:53058    142.250.192.106:443    ESTABLISHED
  TCP    10.20.201.131:53671    40.104.77.34:443       CLOSE_WAIT
  TCP    10.20.201.131:53672    23.212.254.121:443     CLOSE_WAIT
  TCP    10.20.201.131:53673    23.212.254.121:443     CLOSE_WAIT
  TCP    10.20.201.131:53674    23.212.254.121:443     CLOSE_WAIT
  TCP    10.20.201.131:53675    23.37.86.8:443         CLOSE_WAIT
  TCP    10.20.201.131:53676    23.37.86.8:443         CLOSE_WAIT
  TCP    10.20.201.131:53677    23.37.86.8:443         CLOSE_WAIT
  TCP    10.20.201.131:53680    13.107.226.58:443      CLOSE_WAIT
  TCP    10.20.201.131:53681    13.107.9.254:443       CLOSE_WAIT
  TCP    10.20.201.131:53682    13.107.3.254:443       CLOSE_WAIT
  TCP    10.20.201.131:53683    204.79.197.222:443     CLOSE_WAIT
  TCP    10.20.201.131:53730    142.250.207.129:443    ESTABLISHED
  TCP    10.20.201.131:55624    142.250.70.110:443     ESTABLISHED
  TCP    10.20.201.131:56181    57.144.211.32:443      TIME_WAIT
  TCP    10.20.201.131:56182    20.189.173.13:443      TIME_WAIT
  TCP    10.20.201.131:56254    216.58.203.14:443      ESTABLISHED
  TCP    10.20.201.131:57776    57.144.211.32:5222     ESTABLISHED
  TCP    10.20.201.131:58313    52.123.171.43:443      ESTABLISHED
  TCP    10.20.201.131:58318    4.213.25.242:443       ESTABLISHED
```

IV. tracert/traceroute

```
TANMAYI_PES1UG24CS493>tracert google.com

Tracing route to google.com [216.58.203.46]
over a maximum of 30 hops:

  1    19 ms     *        *     10.20.200.1
  2     *        *        *     Request timed out.
  3    13 ms    30 ms    10 ms  192.168.254.1
  4    75 ms    21 ms     *     223-30-0-0.lan.sify.net [223.31.218.222]
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *       28 ms  100.70.136.109
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    18 ms    14 ms    20 ms  100.70.136.25
 11    25 ms    13 ms    11 ms  1.7.105.125
 12   901 ms     *       13 ms  142.251.227.215
 13    14 ms    11 ms    10 ms  172.253.70.166
 14    35 ms    24 ms    39 ms  216.239.49.131
 15    33 ms    27 ms    24 ms  192.178.254.234
 16   394 ms    25 ms    45 ms  192.178.110.109
 17    70 ms   509 ms    29 ms  216.239.54.147
 18   697 ms    36 ms    60 ms  bom12s05-in-f14.1e100.net [216.58.203.46]

Trace complete.
```

V. nslookup

```
TANMAYI_PES1UG24CS493>nslookup google.com
Server:  UnKnown
Address:  192.168.254.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:818::200e
          142.250.195.46
```

VI. arp

```
TANMAYI_PES1UG24CS493>arp -a

Interface: 10.20.201.131 --- 0x6
  Internet Address      Physical Address      Type
  10.20.200.1           14-58-d0-d4-86-00     dynamic
  10.20.200.127         38-7a-0e-a9-df-e4     dynamic
  10.20.200.134         d0-39-57-9e-f3-1f     dynamic
  10.20.200.137         a8-e2-91-84-94-9e     dynamic
  10.20.200.138         9c-fc-e8-89-27-6d     dynamic
  10.20.200.159         14-18-c3-b2-51-f6     dynamic
  10.20.200.160         8c-c6-81-2c-b2-3a     dynamic
  10.20.200.165         d0-ab-d5-71-cb-9a     dynamic
  10.20.200.167         6c-6a-77-95-4e-5d     dynamic
  10.20.200.168         4c-49-6c-3b-3e-a4     dynamic
  10.20.200.170         d4-e9-8a-01-0f-18     dynamic
  10.20.200.177         f8-fe-5e-8a-75-d2     dynamic
  10.20.200.180         70-15-fb-16-8c-3f     dynamic
  10.20.200.181         dc-97-ba-97-74-b7     dynamic
  10.20.200.182         cc-47-40-ec-a4-b9     dynamic
  10.20.200.185         dc-45-46-55-20-da     dynamic
  10.20.200.186         3c-0a-f3-57-6c-c3     dynamic
  10.20.200.187         16-b0-67-67-39-d7     dynamic
  10.20.200.200         60-ff-9e-7a-e2-56     dynamic
  10.20.200.202         de-a2-cc-3f-7f-16     dynamic
  10.20.200.205         f8-3d-c6-d6-1d-44     dynamic
  10.20.200.210         a0-b3-39-4f-62-99     dynamic
  10.20.200.218         30-89-4a-d2-92-a4     dynamic
  10.20.200.221         4c-49-6c-e9-2b-c2     dynamic
  10.20.200.230         64-d6-9a-e9-bd-fd     dynamic
  10.20.200.231         0a-d7-a5-72-20-1c     dynamic
  10.20.200.238         dc-45-46-ca-50-44     dynamic
  10.20.200.240         c4-3d-1a-96-e1-55     dynamic
  10.20.200.246         30-f6-ef-e9-38-42     dynamic
  10.20.200.247         28-2e-89-17-d4-06     dynamic
  10.20.201.5           20-0b-74-1f-de-e5     dynamic
  10.20.201.7           1c-ce-51-ad-49-b6     dynamic
  10.20.201.20          bc-38-98-9d-7a-d3     dynamic
  10.20.201.21          46-cf-f6-41-b9-6f     dynamic
  10.20.201.22          78-46-5c-ad-d1-23     dynamic
  10.20.201.41          40-1c-83-dc-49-75     dynamic
```

VII. netsh

```
TANMAYI_PES1UG24CS493>netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
---------------------------------
    <None>

User profiles
-------------
    All User Profile     : Element BLK(LAPTOP-ONLY)
    All User Profile     : dontrob
    All User Profile     : Platform Nine and Three Routers
    All User Profile     : Olive
    All User Profile     : D-Link_DIR-816
    All User Profile     : Tanmayi
    All User Profile     : D-Link_DIR-816_5G
    All User Profile     : jivin
    All User Profile     : GJBC_Cafeteria
    All User Profile     : GJBC_Library
    All User Profile     : venkatesh tp-link
    All User Profile     : Sri@5GHz
    All User Profile     : realme C33 2023
    All User Profile     : chandu's Galaxy M32
    All User Profile     : Vibhav_5G
    All User Profile     : Vibhav
    All User Profile     : Venky_5G
    All User Profile     : Sarojinic_EXT
    All User Profile     : Sarojinic_5G
    All User Profile     : Redmi Note 11
    All User Profile     : Phone_1_4551
    All User Profile     : Galaxy M12 BE69
    All User Profile     : Gadjetfix tech_5G
    All User Profile     : CHERUKURI'S...5G
    All User Profile     : Airtel_ginj_2417
    All User Profile     : Vishnuinfotech
```

VIII. getmac

```
TANMAYI_PES1UG24CS493>getmac /v

Connection Name Network Adapter Physical Address    Transport Name
=============== ============== ================== ===========================================================
Wi-Fi           Intel(R) Wi-Fi  3C-21-9C-7C-FD-9E   \Device\Tcpip_{1C22ED95-2553-4E31-8F30-CE344F5007B2}
Bluetooth Netwo Bluetooth Devic 3C-21-9C-7C-FD-A2   Media disconnected
Ethernet        VirtualBox Host 0A-00-27-00-00-28   \Device\Tcpip_{6DD54260-651F-4FFD-B93B-2BF94B2472A8}
```

IX. tasklist

```
TANMAYI_PES1UG24CS493>tasklist /svc

Image Name                     PID Services
========================= ======== ============================================
System Idle Process              0 N/A
System                           4 N/A
Secure System                  108 N/A
Registry                       148 N/A
smss.exe                       600 N/A
csrss.exe                      936 N/A
wininit.exe                    348 N/A
csrss.exe                      628 N/A
winlogon.exe                  1056 N/A
services.exe                  1068 N/A
LsaIso.exe                    1124 N/A
lsass.exe                     1140 KeyIso, SamSs, VaultSvc
svchost.exe                   1268 BrokerInfrastructure, DcomLaunch, PlugPlay,
                                   Power, SystemEventsBroker
fontdrvhost.exe               1292 N/A
fontdrvhost.exe               1296 N/A
WUDFHost.exe                  1348 N/A
svchost.exe                   1432 RpcEptMapper, RpcSs
svchost.exe                   1480 LSM
svchost.exe                   1580 HvHost
svchost.exe                   1596 CryptSvc
svchost.exe                   1604 nsi
svchost.exe                   1620 BTAGService
svchost.exe                   1628 BthAvctpSvc
svchost.exe                   1636 bthserv
svchost.exe                   1756 NcbService
svchost.exe                   1764 TimeBrokerSvc
svchost.exe                   1776 Wcmsvc
svchost.exe                   1904 AppIDSvc
svchost.exe                   1980 ProfSvc
IntelCpHDCPSvc.exe            1992 cplspcon
svchost.exe                   2044 DisplayEnhancementService
svchost.exe                    940 netprofm
svchost.exe                   2064 EventSystem
svchost.exe                   2084 NgcSvc
svchost.exe                   2148 UserManager
```

X. route

```
TANMAYI_PES1UG24CS493>route print
===========================================================================
Interface List
 40...0a 00 27 00 00 28 ......VirtualBox Host-Only Ethernet Adapter
  9...3c 21 9c 7c fd 9f ......Microsoft Wi-Fi Direct Virtual Adapter
  5...3e 21 9c 7c fd 9e ......Microsoft Wi-Fi Direct Virtual Adapter #2
  6...3c 21 9c 7c fd 9e ......Intel(R) Wi-Fi 6 AX201 160MHz
 11...3c 21 9c 7c fd a2 ......Bluetooth Device (Personal Area Network)
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      10.20.200.1    10.20.201.131     30
      10.20.200.0    255.255.248.0         On-link     10.20.201.131    286
    10.20.201.131  255.255.255.255         On-link     10.20.201.131    286
    10.20.207.255  255.255.255.255         On-link     10.20.201.131    286
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     192.168.56.0    255.255.255.0         On-link      192.168.56.1    330
     192.168.56.1  255.255.255.255         On-link      192.168.56.1    330
   192.168.56.255  255.255.255.255         On-link      192.168.56.1    330
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     10.20.201.131    286
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    330
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     10.20.201.131    286
  255.255.255.255  255.255.255.255         On-link      192.168.56.1    330
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
```

XI. whoami

```
TANMAYI_PES1UG24CS493>whoami
desktop-ncmb1n9\tanma
```

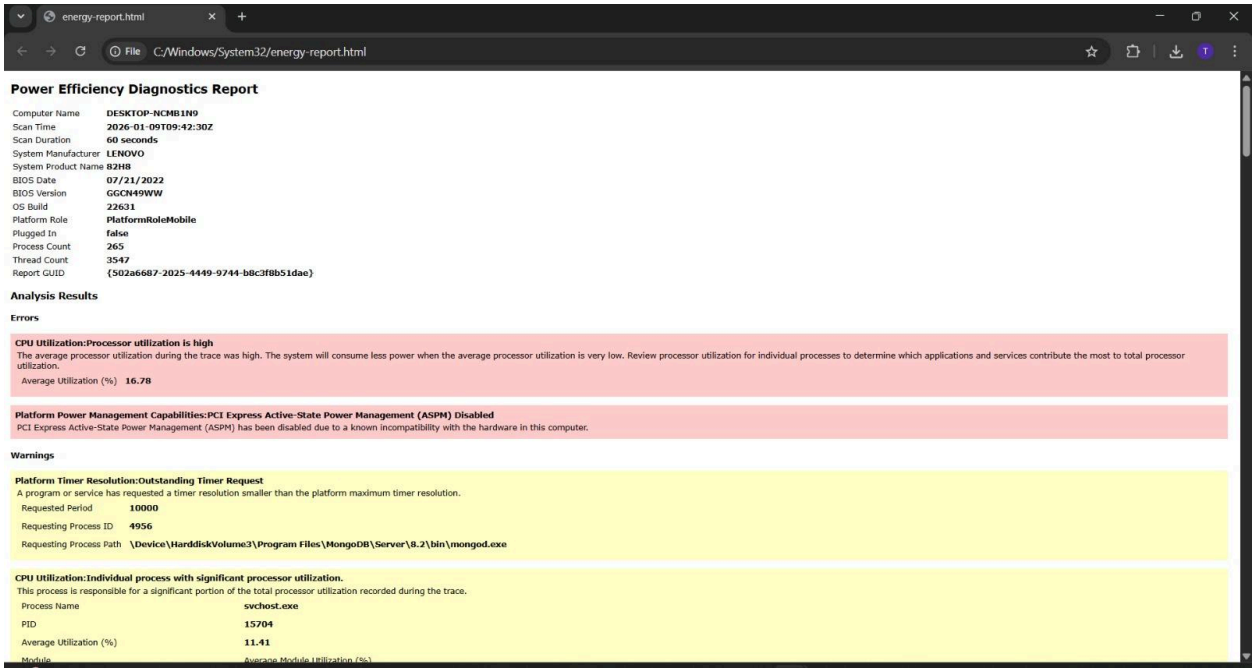
XII. powercfg/pmset -g/upower -d

```
TANMAYI_PES1UG24CS493>powercfg /energy
Enabling tracing for 60 seconds...
Observing system behavior...
Analyzing trace data...
Analysis complete.

Energy efficiency problems were found.

2 Errors
9 Warnings
59 Informational

See C:\Windows\System32\energy-report.html for more details.
```



XIII. systeminfo

```
TANMAYI_PES1UG24CS493>systeminfo

Host Name:                 DESKTOP-NCMB1N9
OS Name:                   Microsoft Windows 11 Home
OS Version:                10.0.22631 N/A Build 22631
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          hymanalluri13@gmail.com
Registered Organization:
Product ID:                00342-20932-45151-AAOEM
Original Install Date:     07/06/2025, 09:08:27
System Boot Time:          06/01/2026, 23:26:20
System Manufacturer:       LENOVO
System Model:              82H8
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 140 Stepping 1 GenuineIntel ~2419 Mhz
BIOS Version:              LENOVO GGCN49WW, 21/07/2022
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              00004009
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     7,975 MB
Available Physical Memory: 1,010 MB
Virtual Memory: Max Size:  25,383 MB
Virtual Memory: Available: 13,028 MB
Virtual Memory: In Use:    12,355 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\DESKTOP-NCMB1N9
Hotfix(s):                 5 Hotfix(s) Installed.
                           [01]: KB5066133
                           [02]: KB5027397
                           [03]: KB5041655
                           [04]: KB5058405
                           [05]: KB5058528
Network Card(s):           3 NIC(s) Installed.
                           [01]: Intel(R) Wi-Fi 6 AX201 160MHz
```

XIV. sc

```
TANMAYI_PES1UG24CS493>sc query dnscache

SERVICE_NAME: dnscache
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE              : 4  RUNNING
                                (NOT_STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE    : 0  (0x0)
        SERVICE_EXIT_CODE  : 0  (0x0)
        CHECKPOINT         : 0x0
        WAIT_HINT          : 0x0
```

XV. telnet

```
TANMAYI_PES1UG24CS493>telnet example.com 80
'telnet' is not recognized as an internal or external command,
operable program or batch file.
```

XVI. fsutil/diskutil

```
TANMAYI_PES1UG24CS493>fsutil behaviour set disable8dot3 1
behaviour is an invalid parameter.
---- Commands Supported ----

8dot3name         8dot3name management
behavior          Control file system behavior
bypassIo          BypassIo management
dax               Dax volume management
devdrv            Developer volume management
dirty             Manage volume dirty bit
file              File specific commands
fsInfo            File system information
hardlink          Hardlink management
objectID          Object ID management
quota             Quota management
repair            Self healing management
reparsePoint      Reparse point management
storageReserve    Storage Reserve management
resource          Transactional Resource Manager management
sparse            Sparse file control
tiering           Storage tiering property management
trace             File system trace management
transaction       Transaction management
usn               USN management
volume            Volume management
wim               Transparent wim hosting management
```

XVII. wmic

```
TANMAYI_PES1UG24CS493>wmic nic get name, macaddress
MACAddress         Name
                   Microsoft Kernel Debug Network Adapter
3C:21:9C:7C:FD:9E  Intel(R) Wi-Fi 6 AX201 160MHz
3C:21:9C:7C:FD:9F  Microsoft Wi-Fi Direct Virtual Adapter
                   WAN Miniport (SSTP)
                   WAN Miniport (IKEv2)
                   WAN Miniport (L2TP)
                   WAN Miniport (PPTP)
                   WAN Miniport (PPPOE)
BA:8D:20:52:41:53  WAN Miniport (IP)
BC:04:20:52:41:53  WAN Miniport (IPv6)
BE:15:20:52:41:53  WAN Miniport (Network Monitor)
3E:21:9C:7C:FD:9E  Microsoft Wi-Fi Direct Virtual Adapter #2
3C:21:9C:7C:FD:A2  Bluetooth Device (Personal Area Network)
0A:00:27:00:00:28  VirtualBox Host-Only Ethernet Adapter
```

XVIII. net user

```
TANMAYI_PES1UG24CS493>net user Administrator
User name                    Administrator
Full Name
Comment                      Built-in account for administering the computer/domain
User's comment
Country/region code          000 (System Default)
Account active               No
Account expires              Never

Password last set            09/01/2026 15:39:38
Password expires             Never
Password changeable          09/01/2026 15:39:38
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships      *Administrators
Global Group memberships     *None
The command completed successfully.
```